[INCH-POUND]
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June 16, 1998
SUPERSEDING
KKK-E-2810
March 8, 1993

COMMERCIAL ITEM DESCRIPTION

EXCAVATOR, MULTIPURPOSE, WHEEL-UNDERCARRIAGE DIESEL-ENGINE-DRIVEN

The General Services Administration has authorized the use of this commercial item description for all Federal agencies.

1. SCOPE. This commercial item description (CID) covers a hydraulic powered excavator. The excavator shall consist of two-axle undercarriage, revolving upperstructure, diesel engine, and a hoe-type boom. The excavator is intended for use in excavating and loading loose gravel, sand, or soil.

2. SALIENT CHARACTERISTICS.

2.1 Description. The excavator consists of a revolving upperstructure. This upperstructure shall consist of an operator's cab, all operational controls, hydraulic system, and the boom assembly. The diesel engine shall power the hydraulic system and provide motivation for all functions including travel function. The excavator shall have a rated lift capacity of not less than 9,300 pounds (lb) at 15-foot (ft) (4 222 kilogram (kg) at 4.57 metre (m)) radius at ground level.

2.2 Standard commercial product. The excavator shall, as a minimum, be in accordance with the requirements of this CID and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this CID but which are a part of the manufacturer's standard commercial product, shall be included in the excavator being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements or manufacturer's catalogs, or brochures. Additional options may be added at time of issuance of contract or purchase as required by purchasing agency.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data which may improve this document should be sent to: Officer In Charge (Code 15E2), Seabee Logistics Center, 4111 San Pedro St., Port Hueneme, CA 93043-4410, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 3805

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2.3 Interchangeability. All units of the same classification furnished with similar options under a specific contract shall be identical to the extent necessary to ensure interchangeability of component parts, assemblies, accessories, and spare parts.

2.4 Safety. All rotating or reciprocating parts, and all parts subject to high operational temperatures, that are of such nature or are located as to be or become a hazard to the operating or attending personnel, shall be guarded or insulated. Walking or working surfaces and platforms shall be of an anti-skid type. Handholds or steps or both shall be provided to allow safe entrance and exit from the operator's cab.

2.4.1 Noise. When the noise level, tested in accordance with MIL-STD-1474, is greater than 85 decibels (db(A)), a caution plate shall be permanently attached in a conspicuously protected location. The plate shall be clearly visible and legible to all personnel exposed to excessive noise levels. The caution plate shall read: CAUTION - HEARING PROTECTION REQUIRED. The plate shall have a yellow background with black lettering and shall be made of corrosion-resistant material.

2.4.2 Electrical warning decal. An electrical warning decal, in accordance with Occupational Safety and Health Administration (OSHA) requirements, cautioning the operator about operating the excavator near electrical power lines, shall be provided.

2.4.3 Strike hazard. The rear surfaces of the excavator shall be alternating diagonal stripes, black and yellow in color. The stripes shall be not less than 18 inches (457 millimetre (mm)) in height from lower edge of upperstructure. The stripes shall alternate in equal widths across rear of upperstructure. The stripes shall lap around each rear corner and forward not less than 1-ft (305 mm) distance along each side.

2.5 Environmental conditions. The excavator shall start and maintain constant performance characteristics as specified herein under any of the following conditions or combination of conditions:

a. Temperature conditions. In any ambient temperature from - 20 degrees Fahrenheit (°F) to 110 °F (- 29 degrees Celsius (°C) to 43 °C), except as specified in 2.6b below.

b. Elevation conditions. At any elevation from sea level (barometric pressure of 29.92 inches mercury (101.32 kilopascal (kPa)) and ambient temperature not greater than 100 °F (38 °C) to 5,000 ft (1 524 m) (barometric pressure 24.89 inches of mercury (84.27 kPa)) and a temperature of 100 °F (38 °C).

2.6 Upper structure.

2.6.1 Cab. The cab shall be fabricated of sheet metal or sheet fiberglass and shall have sufficient windows and glazed doors to permit 200 degrees horizontal and 90 degrees vertical vision, when measured from top rear area of the operator's seat. Visual access from the operator's position to ground level, shall be not greater than 5 ft (1.5 m) forward of tires. All glass shall be of the safety

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type. Cab door(s), whether of the sliding or swinging type, shall be adequately restrained from accidentally opening or closing while the excavator is traveling or operating. An adjustable seat for the operator shall be provided.

2.6.2 Swing lock mechanism. A means capable of restraining the upperstructure in any desired position, with respect to the undercarriage, shall be provided. The swing lock or brake mechanism shall be in accordance with PCSA Standard No. 5, and shall be located within easy reach of the operator.

2.7 Operating controls. All functions shall be in accordance with SAE J 1177. All brakes shall be capable of being easily locked or unlocked. All operating controls shall be identified in accordance with SAE J 1362 recommended practice.

2.8 Hydraulic system and operation. High pressure hydraulic hoses and fittings shall be capable of withstanding a bursting test pressure of four times the working pressure. High pressure hydraulic hoses shall have the physical qualities equal to or conforming to SAE J 517. The following functions shall be hydraulically operated and controlled:

a. Boom and bucket operation.
b. Revolving of upperstructure.
c. Operation of optional attachments and accessories (see 2.9).
d. Hydrostatic drive function (see 2.11.1).

2.9 Boom and attachments.

2.9.1 Boom. The excavator shall be supplied with a hoe-type boom. The boom shall have the following characteristics and capabilities, when measured in accordance with SAE J 1193 and SAE J 1179 recommended practice:

a. Maximum radius at ground line (AA) not less than 30 ft (9.1 m).
b. Maximum digging depth 8 ft (2.4 m) flat floor (AB) not less than 18 ft (5.5 m).
c. Dumping height (BD) not less than 15 ft (4.57 m).
d. Swing shall be continuous.
e. Digging through a continuous range around the excavator not less than 360 degrees.
f. Bucket digging force not less than 22,000 lb (98.0 kilonewton (kN).
g. Rated arm force not less than 15,500 lb (69.0 kN).

2.9.2 Attachments.

2.9.2.1 Excavating bucket. The following excavating buckets shall be furnished as specified (see 6.2). All dimensions are nominal, within 2 inches (51 mm) total, capacity nominal within 10 percent.

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	Minimum capacity (struck) cubic yard (yd^3)	cubic metre (m^3)
a. Bucket, excavating, 24-inch (610 mm) cutting width	0.56	(0.43)
b. Bucket, excavating, 36-inch (914 mm) cutting width	0.88	(0.67)
c. Bucket, excavating, 48-inch (1219 mm) cutting width	1.00	(0.76)

2.9.2.2 Hydraulic impact hammer. When specified (see 6.2), a hydraulic impact hammer shall be provided with the excavator. The hydraulic impact hammer shall operate from the excavator's hydraulic system. The impact hammer and excavator shall be equipped with a quick disconnect system to allow rapid hookup or removal of the hammer. The necessary pins, bolts, and other necessary hardware shall be furnished by the manufacturer to enable hookup of the impact hammer to the excavator arm. The quick disconnect shall be designed for one person operation. Tools required shall be furnished with each unit. The impact hammer shall meet the following requirements:

a. Energy output per minute	not less than 625,000 foot-pounds (847.4 kilojoules (kJ)).
b. Impact energy class	not less than 1,500 lb (680 kg).
c. Blows per minute	not less than 500.

The hydraulic impact hammer shall be furnished with not less than following tools:

d. Moil point	3 each.
e. Chisel	1 each.
f. Line cutter	1 each.
g. Spade	1 each.
h. Tamper plate assembly	1 each (not less than 144 square inch (0.9 square metre) face area).

2.9.2.3 Hydraulic vibratory compactor. When specified (see 6.2), the excavator shall be equipped with a hydraulic vibratory compactor. The hydraulic vibratory compactor shall have a plate area of not less than 9 square feet (0.84 square metre). The hydraulic vibratory compactor shall be capable of operating from the excavator's hydraulic system. The hydraulic vibratory compactor and excavator shall be equipped with quick disconnect systems to allow rapid hookup or removal. The necessary pins, adapters, and other necessary hardware shall be furnished by the manufacturer to enable hookup of the hydraulic vibratory compactor to the excavator's arm. The excavator shall be capable of generating a rated impulse force of 20,000 lb (89 kN) at 2,000 cycles per minute. The excavator shall not have loss of stick or boom function while operating the compactor plate.

2.10 Rated lifting capacity. Rated lift capacities shall be in conformance with SAE J 1097. The lifting capacity shall be met without the use of stabilizers in any arc 360 degrees around the excavator. Manufacturer's load chart shall specify the rated lift capacity over side.

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2.11 Undercarriage. The undercarriage for the excavator shall be a heavy-duty frame, with not less than two axles or drive units, supported by pneumatic rubber tires.

2.11.1 Transmission. Engine power shall be transmitted through either a torque converter with a transmission, or a full hydrostatic drive system with infinitely variable forward and reverse speeds throughout range capability. When a transmission is furnished, not less than three forward and at least one reverse speeds shall be provided. The transmission shall be capable of operator selected shifting under full power load in either direction.

2.11.1.1 Drive train. The drive train shall provide power to not less than four drive wheels and obtain not less than 40 percent of the vehicle's weight in a drawbar pull, when on level, dry, brushed concrete. When differential-type transmissions are part of the excavator's assemblage, a limited-slip device shall be included in the drive train. Wheel drive components shall be designed to accept the maximum deliverable torque load by the drive train without adverse effects under all pulling conditions. When furnished with hydrostatic drive, units shall deliver not less than 19,000 lb (84.5 kN) of traction force to each wheel.

2.11.2 Speed. The excavator shall be capable of traveling forward on level pavement not less than 18 miles per hour (mph) (29 kilometre per hour (km/h)).

2.11.3 Gradeability. The excavator shall be capable of negotiating a 30 percent grade on dry concrete or asphalt at a speed of not less than 1.2 mph (1.9 km/h).

2.12 Outriggers. The outriggers shall be hydraulically powered, operated from the revolving upperstructure cab (see 2.6.1). The outrigger beam(s) shall be equipped with floats or outrigger pads. Outriggers shall be individually controlled. Outriggers shall be capable of being adjusted to not less than 4 inches (102 mm) below ground level and 3 inches (76 mm) above ground level to compensate for uneven terrain. A device or system to accurately determine when the excavator is level shall be provided.

2.13 Brakes. The excavator, other than hydraulically driven types, shall be equipped with the manufacturer's standard braking system, in conformance with SAE J 1152. Hydraulically driven types shall comply with the applicable stopping performance, holding performance, and parking system performance criteria of SAE J 1152.

2.14 Steering. The manufacturer's standard steering system shall be provided. Hydraulic skid steering is acceptable.

2.15 Lighting. The excavator shall be equipped with not less than the following lights:

a. Front and rear work light(s).
b. Boom mounted work light(s).
c. Two dual stop and taillight(s), and directional turn signals.
d. Interior cab light(s).

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2.16 Engine. When specified (see 6.2), the battery furnished in the excavator shall be of the maintenance-free type having maintenance-free characteristics. The engine shall be equipped with the following:

a. Hour meter.
b. Cooling liquid temperature indicator.
c. Lubricating oil pressure indicator.
d. Liquid fuel gage.
e. Fuel tank with sufficient capacity for 8 hours normal operation.

2.16.1 Cold starting system. The excavator diesel engine shall be equipped with an engine priming system, glow plug(s), or intake manifold preheater design for cold temperature engine starting. When an ether priming system is furnished, it shall be of the measured shot type with a reservoir of not less than 12 fluid ounces (355 millilitre).

2.17 Tires. Tires shall be of the tubeless type, and shall have individual rated load-carrying capacities conforming to Tire and Rim Association (TRA) recommendations. When specified (see 6.2), tires shall be foam-filled.

2.18 Dozer blade. When specified (see 6.2), manufacturer's standard hydraulic operated dozer blade assembly shall be provided. The dozer blade shall operate from the excavator's hydraulic system. The hydraulic dozer blade assembly shall be equipped with quick disconnect system to allow rapid hookup or removal.

2.19 Lifting and tiedown attachments. When specified (see 6.2), the excavator shall be equipped with lifting and tiedown attachments that meet requirements of MIL-STD-209. A nonferrous transportation plate shall be provided and mechanically attached to the excavator. Transportation plates shall be inscribed with a diagram showing the lifting attachments and lifting slings, the capacity of each attachment, and the required length and size of each sling cable. A silhouette of the item furnished showing the center of gravity shall be provided on the transportation plate. Tiedown attachments may be identified by stenciling or other suitable marking. Tiedown marking shall clearly indicate that the attachments are intended for the tiedown of the excavator on the carrier when shipped.

2.20 Instruction plates. The excavator shall be equipped with instruction plates suitably located, describing any special or important procedures to be followed in operating and servicing the equipment. Plates shall be of a material which will last and remain legible for the life of the equipment. Plates shall be securely affixed to the equipment with nonferrous screws, bolts, or rivets. Heavy-duty decals made of material which will last and remain legible for the life of the equipment are acceptable.

2.21 Cleaning, treatment, and painting. Surfaces normally painted in good commercial practice shall be cleaned, treated, and painted as specified herein. The color of the finish coat shall be as specified (see 6.2). Surfaces to be painted shall be cleaned and dried to ensure that they are free from contaminants such as oil, grease, welding slag and spatter, loose mill scale, water, dirt,

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corrosion product, or any other contaminating substances. As soon as practicable after cleaning, and before any corrosion product or other contamination can result, the surfaces shall be prepared or treated to ensure the adhesion of the coating system. The painting shall consist of at least one coat of primer and one finish coat. The primer shall be applied to a clean, dry surface as soon as practicable after cleaning and treating. Painting shall be with the manufacturer's current materials according to manufacturer's current processes and the total dry film thickness shall be not less than 2.5 mils (0.06 mm) over the entire surface. The paint shall be free from runs, sags, orange peel, or other defects.

2.22 Lubrication. Means for lubrication shall be in accordance with the manufacturer's standard practice. The lubricating points shall be easily visible and accessible. Hydraulic lubrication fittings shall be in accordance with SAE J 534. Where use of high-pressure lubricating equipment, 1,000 pound-force per square inch (6 895 kPa) or higher, will damage grease seals or other parts, a suitable warning shall be affixed to the equipment in a conspicuous location. The unit shall be lubricated prior to delivery with type of lubricant specified in the operator's manual and grade of lubricant recommended for ambient temperature at the delivery point. The unit shall be conspicuously tagged to identify the lubricants and their temperature range.

2.23 Servicing and adjusting. Prior to acceptance of the excavator by the Government, the contractor shall service and adjust the excavator for immediate operational use as required in the operator's manual. The servicing and adjusting shall include at least the following:

a. Inflation of all tires.
b. Adjustment of brakes.
c. Proper functioning of all lighting and electrical systems.
d. Wheel alignment (when required).
e. Adjustment of engine to include tune-up (when required).
f. Complete lubrication with grades of lubricants recommended for ambient temperature at the delivery point.
g. Cooling system filled to capacity with a clean solution of equal parts by volume of water and antifreeze (ethylene glycol).

2.24 Rail transport. When specified (see 6.2), the excavator shall pass the MIL-STD-810 rail impact test without damage or permanent deformation.

2.25 Air transportability configuration. When specified (see 6.2), the excavator shall be transportable by C-130, C-141, C-5, and C-17 aircraft in a roll-on/roll-off configuration. Refer to MIL-STD-1791 for dimensional and weight guidance for meeting this requirement.

2.26 Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within specified tolerances using conversion tables contained in the latest version of ASTM SI-10, and all other requirements of this commercial item description including form, fit and function are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch-pound units, a request should be made to the contracting officer to determine

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if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

3. REGULATORY REQUIREMENTS.

3.1 Materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4. QUALITY ASSURANCE PROVISIONS.

4.1 Product conformance. The products provided shall meet the salient characteristics of this commercial item description, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial market. The government reserves the right to require proof of such conformance.

5. PACKAGING. The preservation, packing, and marking shall be as specified in the contract or order.

6. NOTES.

6.1 Source of documents.

6.1.1 Copies of military standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.

6.1.2 The Federal Acquisition Regulation (FAR) and OSHA standards may be obtained from the Superintendent of Documents, U.S. Printing Office, Washington, DC 20402.

6.1.3 ASTM Standard is available from the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.

6.1.4 PCSA Standards are available from the Power Crane and Shovel Association, Marina Plaza, Suite 1700, 111 E. Wisconsin Avenue, Milwaukee, WI 53202-4879.

6.1.5 SAE Standards are available from the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.

6.1.6 TRA Yearbook Standards may be obtained from Tire and Rim Association Inc., 175 Montrose West Avenue, Suite 150, Copley, OH 44321.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this CID.
b. List of attachments and accessories for each excavator (see 2.9.2.1, 2.9.2.2, and 2.9.2.3).

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c. When battery is of the maintenance-free type (see 2.16).
d. When foam-filled tires are required (see 2.17).
e. When dozer blade assembly is to be provided (see 2.18).
f. When lifting and tiedown attachments are required (see 2.19).
g. Color of finish coat (see 2.21).
h. When rail impact test is required (see 2.24).
i. When air transportability configuration is required (see 2.25).

6.3 Supersession data. This commercial item description replaces Federal Specification KKK-E-2810 dated March 8, 1993.

MILITARY INTERESTS:

Custodians:
Army - AT
Navy - YD1
Air Force - 99

Review Activity:
Army - MT
Air Force - 84

CIVIL AGENCY COORDINATING ACTIVITY:

GSA - FSS

Preparing Activity:
Navy - YD1

(Project 3805-0027)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** A-A-59251	**2. DOCUMENT DATE (YYMMDD)** 980616

3. DOCUMENT TITLE

EXCAVATOR, MULTIPURPOSE, WHEEL-UNDERCHARRIAGE DIESEL-ENGINE-DRIVEN

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON *(if applicable)*	**7.DATE SUBMITTED** (YYMMDD)

8. PREPARING ACTIVITY

a. NAME R. E. PAMPLONA	b. TELEPHONE *Include Area Code)* (1) Commercial 805-982-5843 (2) AUTOVON 551-5843
c. ADDRESS *(Include Zip Code)* OFFICER IN CHARGE, SEABEE LOGISTICS CENTER, CODE 15E2, 4111 SAN PEDRO ST. PORT HUENEME, CA 93043-4410	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** DEFENSE QUALITY AND STANDARDIZATION OFFICE 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22401-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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